# Homework 5

# Electromagnetism - PHYS 715

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**5.3** A magnetic induction **B** in a current-free region in a uniform medium is cylindrically symmetric with components $B_z(\rho, z)$ and $B_\rho(\rho, z)$ and with a known $B_z(0, z)$ on the axis of symmetry. The magnitude of the axial field varies slowly in $z$.

**(a)** Show that near the axis the axial and radial components of magnetic induction are approximately

$$B_z(\rho, z) \approx B_z(0, z) - \left(\frac{\rho^2}{4}\right)\left[\frac{\partial^2 B_z(0, z)}{\partial z^2}\right] + \dots$$

$$B_\rho(\rho, z) \approx -\left(\frac{\rho}{2}\right)\left[\frac{\partial B_z(0, z)}{\partial z}\right] + \left(\frac{\rho^3}{16}\right)\left[\frac{\partial^3 B_z(0, z)}{\partial z^3}\right] + \dots$$

**(b)** What are the magnitudes of the neglected terms, or equivalently what is the criterion defining "near" the axis?

$\vec{B}$ is a static field with no current (i.e. $\partial_t \rho = 0$)

$\therefore$ We have from Maxwell's Equations $\Rightarrow$ $\vec{\nabla} \cdot \vec{B} = 0$

$$\nabla \times \vec{B} = \vec{0}$$

The field is cylinderically symmetric around the z-axis. Therefore, can be written as

$$\vec{B} = B_\rho(\rho, z)\,\hat{\rho} + B_z(\rho, z)\,\hat{z} \quad , \quad B_\phi = 0$$

On the axis, $\rho = 0$ $\Rightarrow$ $\vec{B} = B_z(0, z)\,\hat{z}$

Goal : Find the axial $B_z$ and radial $B_\rho$ components next to the axis $\rho \approx 0$.

a) <u>Power Series Expansion of components with cylinderical symmetry :</u>

We can expand both components using a power series of $\rho$ (centered around $\rho = 0 \Rightarrow$ near axis)

$$B_z(\rho, z) = \sum_{n=0}^{\infty} a_n(z)\,\rho^n$$

$$B_\rho(\rho, z) = \sum_{n=0}^{\infty} b_n(z)\,\rho^n$$

Since the problem is cylinderically symmetric, we must have that :

$B_z(\rho, z) = B_z(-\rho, z)$ $B_z$ is even in $\rho$ since $B_z$ is indifferent to $\rho$ or $-\rho$

$B_\rho(\rho, z) = -B_\rho(-\rho, z)$ $B_\rho$ is odd in $\rho$ since $B_\rho$ is opposite for $\rho$ to $-\rho$.

We therefore then have that :

$B_z(\rho, z) = a_0(z) + a_2(z)\rho^2 + a_4(z)\rho^4 + \dots$ only even terms

$B_\rho(\rho, z) = b_1(z)\rho + b_3(z)\rho^3 + b_5(z)\rho^5 + \dots$ only odd terms

# Connecting components through Maxwell's Equations :

1) Divergence - Free : $\nabla \cdot \vec{B} = 0$

$$\frac{1}{\rho}\partial_\rho(\rho B_\rho) + \cancel{\frac{1}{\rho}\partial_\phi B_\phi}^{0} + \partial_z B_z = 0$$ no $\phi$-dependence (1)

2) Curl - Free : $\vec{\nabla}\times\vec{B} = \vec{0}$

$$\left(\frac{1}{\rho}\partial_\phi B_z - \cancel{\partial_z B_\phi}^{0}\right)\hat{\rho} + \left[\partial_z B_\rho - \partial_\rho B_z\right]\hat{\phi} + \frac{1}{\rho}\left[\cancel{\partial_\rho(\rho B_\phi)}^{0} - \partial_\phi B_\rho\right]\hat{z}$$ no $\phi$-dependence (2)

$$\Rightarrow \partial_z B_\rho = \partial_\rho B_z$$

Apply the $1^{st}$ condition : $\frac{1}{\rho}\partial_\rho(\rho B_\rho) + \partial_z B_z = 0$

$$\rho B_\rho = b_1\rho^2 + b_3\rho^4 + b_5\rho^6 + \cdots$$

$$\partial_\rho(\rho B_\rho) = 2b_1\rho + 4b_3\rho^3 + 6b_5\rho^5 + \cdots$$

$$\frac{1}{\rho}\partial_\rho(\rho B_\rho) = 2b_1 + 4b_3\rho^2 + 6b_5\rho^4 + \cdots$$

$$-\partial_z B_z = -a_0' + a_2'\rho^2 + a_4'\rho^4 + \cdots$$

We now can connect coeffecients $a(z)$ & $b(z)$ by :

$$a_{2n}'(z) = -(2n+2)\, b_{2n+1}(z) \qquad (3)$$

Apply the $2^{nd}$ condition : $\partial_z B_\rho = \partial_\rho B_z$

$$\partial_z B_\rho = b_1'\rho + b_3'\rho^3 + b_5'\rho^5 + \cdots$$

$$\partial_\rho B_z = 2a_2\rho + 4a_4\rho^3 + 5a_5\rho^5 + \cdots$$

We now can connect coeffecients $a(z)$ & $b(z)$ by :

$$b_{2n+1}'(z) = (2n+2)\, a_{2n+2}(z) \qquad (4)$$

Using both (3) and (4). We can now form a recurrence relationshi

Differentiating (3) in $z$ : $a_{2n}''(z) = -(2n+2)\, b_{2n+1}'(z)$

Substitute (4) : $a_{2n}''(z) = -(2n+2)^2\, a_{2n+2}(z) \quad \Rightarrow \quad a_{2n+2}(z) = -\frac{1}{(2n+2)^2}\, \partial_z^2 a_{2n}(z)$ (5)

We note that $a_0(z) = B_z(0,z)$.

## Series Expansion of Axial $B_z$ & Radial $B_\rho$ components :

Axial $B_z$

Recall : $B_z = \sum_{n=0}^{\infty} a_{2n}(z)\rho^{2n} = a_0 + a_2\rho^2 + \dots$

$= a_0 - \frac{1}{2^2}\partial_z^2 a_0 \rho^2 + \dots$ sub in (5)

$\approx B_z(0,z) - \frac{1}{4}\partial_z^2 B_z(0,z)\rho^2$

Radial $B_\rho$

Recall : $B_\rho = \sum_{n=0}^{\infty} b_{2n+1}(z)\rho^{2n+1} = b_1\rho + b_3\rho^3 + \dots$

$= -\frac{a_0'}{2}\rho - \frac{a_2'}{4}\rho^3 + \dots$ sub in (3)

$= -\frac{\rho}{2}\partial_z B_z(0,z) + \frac{\rho^3}{16}\partial_z^3 B_z(0,z)$ sub in (5)

b) Let $L$ be the scale of variation of the field i.e.

$\frac{\partial B_z}{\partial z} = \frac{B_0}{L}$, $\frac{\partial^2 B_z}{\partial z^2} = \frac{B_0}{L^2}$, $\frac{\partial^3 B_z}{\partial z^3} = \frac{B_0}{L^3}$, $\frac{\partial^4 B_z}{\partial z^4} = \frac{B_0}{L^4}$, etc...

Then, since for all power series terms of both axial $B_z$ and radial $B_\rho$,

we have $\frac{\partial^i B_z(0,z)}{\partial z^i}\cdot \rho^i$ power of $\rho$ is the same as derivative counter.

$\Rightarrow \quad B_0 \frac{\rho^i}{L^i}$

For large terms, we want $\frac{\rho^i}{L^i}$ to be negligible i.e. $\rho \ll L$ which is exactly the problem we are solving. $\Rightarrow$ Near the axis $\rho$ is small compared to scale $L$.

**5.4 (a)** Use the results of Problem 5.3 and Worked Problem 5.1 to find the axial and radial components of magnetic induction in the *central region* ($|z| << L/2$) of a long uniform solenoid of radius $a$ and ends at $z = \pm L/2$, including the value of $B_z$ just inside the coil ($\rho = a^-$).

**(b)** Use Ampère's law to show that the longitudinal magnetic induction just outside the coil is approximately

$$B_z(\rho = a^+,\ z) \approx -\left(\frac{2\mu_0 N I a^2}{L^2}\right)\left(1 + \frac{12z^2}{L^2} - \frac{9a^2}{L^2} + \ldots\right)$$

For $L >> a$, the field outside is negligible compared to inside. How does this axial component compare in size to the azimuthal component of Problem 5.2b?

**(c)** Show that at the end of the solenoid the magnetic induction near the axis has components

$$B_z \simeq \frac{\mu_0 NI}{2},\quad B_\rho \simeq \pm\frac{\mu_0 NI}{4}\left(\frac{\rho}{a}\right)$$

We are trying to find the magnetic induction $\vec{B}$ in the central region of a solenoid.

Radius : $a$

Solenoid Ends : $z = \pm L/2$

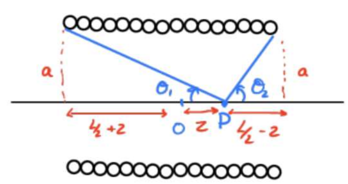
Central Region : $|z| << L/2$

Long : $a << L$.

a) From worked problem 5.1,

the magnetic induction $\vec{B}_z$ on the axis of a solenoid with length $L$ & $N$ turns is :

$$B_z(\rho=0, z) = \frac{\mu_0 NI}{2}(\cos\theta_1 + \cos\theta_2)$$




$$= \frac{\mu_0 NI}{2}\left(\frac{z + L/2}{\sqrt{a^2 + (z + L/2)^2}} - \frac{z - L/2}{\sqrt{a^2 + (z - L/2)^2}}\right) \quad (1)$$

From problem 5.3, we have that the axial & radial component depend on $B_z(\rho=0, z)$ by :-

$$B_z(\rho, z) = B_z(0, z) - \frac{\rho^2}{4}\partial_z^2 B_z(0, z)$$

$$B_\rho(\rho, z) = -\frac{\rho}{2}\partial_z B_z(0, z) + \frac{\rho^3}{16}\partial_z^3 B_z(0, z)$$

$B_z(0, z)$ is an unruly function and difficult to find the derivatives for. However, we can exploit our conditions!

## Taylor Series Expansion for $B_z(0,z)$:

We know from 5.3 that $B_z$ is even. Therefore, the first 2 terms of the Taylor Series Expansion at the origin (since we are looking at $z$ in the central region $z \approx 0$)

$$B_z(0,z) = B_z(0,0) + \frac{1}{2} B_z''(0,0) z^2$$

## Approximating the coeffecients $B_z(0,0)$ & $B_z''(0,0)$:

$B_z(0,0)$

$$B_z(0,0) = \frac{\mu_0 NI}{2}\left[\frac{L/2}{\sqrt{a^2+(L/2)^2}} + \frac{L/2}{\sqrt{a^2+(L/2)^2}}\right] = \mu_0 NI\left[\frac{L}{\sqrt{4a^2+L^2}}\right]$$

For a long solenoid ($a \ll L$), we expand in small parameter $a/L$.

$$B_z(0,0) = \mu_0 NI\left[\frac{L}{\sqrt{L^2+4a^2}}\right] = \mu_0 NI\left[\frac{1}{\sqrt{1+4\frac{a^2}{L^2}}}\right]$$

$$= \mu_0 NI\left(1 - \frac{2a^2}{L^2} + \frac{6a^4}{L^4} + \dots\right)$$

Using binomial expansion $(1+x)^{-1/2} = 1 - \frac{1}{2}x + \frac{3}{8}x^2 + \dots$

$B_z''(0,0) \Rightarrow$ we need to differentiate twice.

$$B_z(0,z) = \frac{\mu_0 NI}{2}\left(\frac{z+L/2}{\sqrt{a^2+(z+L/2)^2}} - \frac{z-L/2}{\sqrt{a^2+(z-L/2)^2}}\right)$$

* we see that $B_z$ is of the form $f(u) = \frac{u}{\sqrt{a^2+u^2}} = \left(\frac{a^2}{u^2}+1\right)^{-1/2}$.

We find $f''(u)$ & make a substitute of variables to get $B_z''(0,z)$

$$f(u) = \left(\frac{a^2}{u^2}+1\right)^{-1/2}$$

$$f'(u) = -\frac{1}{2}\left(\frac{a^2}{u^2}+1\right)^{-3/2} \cdot -2\frac{a^2}{u^3} = \frac{a^2}{u^3}\left(\frac{a^2}{u^2}+1\right)^{-3/2}$$

$$f''(u) = \frac{a^2}{u^3}(-3/2)\left(\frac{a^2}{u^2}+1\right)^{-5/2}\left(-2\frac{a^2}{u^3}\right) - 3\frac{a^2}{u^4}\left(\frac{a^2}{u^2}+1\right)^{-3/2}$$

$$= \frac{3\frac{a^4}{u^6} - 3\frac{a^2}{u^4}\left(\frac{a^2}{u^2}+1\right)}{\left(\frac{a^2}{u^2}+1\right)^{5/2}} = -\frac{3\frac{a^2}{u^4}}{\left(\frac{a^2}{u^2}+1\right)^{5/2}} = -\frac{3a^2u}{(a^2+u^2)^{5/2}}$$

$$\Rightarrow B_z''(0,0) = \frac{\mu_0 NI}{2}\left[f''(L/2) - f''(-L/2)\right]$$

$$B_z''(0,0) = \frac{\mu_0 NI}{2}\left[\frac{-3a^2L}{[a^2+(L/2)^2]^{5/2}}\right] = -\frac{48\mu_0 NI a^2 L}{(4a^2+L^2)^{5/2}}$$

$\Rightarrow$ Expanding $B_z''$ for $(a \ll L)$ to get polynomial in $a/L$.

$$B_z''(0,0) = -48\mu_0 NI\ a^2 L^{-4}\left(1+4\frac{a^2}{L^2}\right)^{-5/2}$$

$$= -48\mu_0 NI\ a^2 L^{-4}\left(1-10\frac{a^2}{L^2}\right)$$

$$\approx -48\mu_0 NI\ \frac{a^2}{L^4}$$

Using binomial expansion

$(1+x)^{-5/2} = 1 - \frac{5}{2}x + \ldots$

Finally! Finding the components :

Axial Component $B_z$ :

Recall : $B_z(\rho,z) \approx B_z(0,z) - \frac{\rho^2}{4}\partial_z^2 B_z(0,z)$

$$\approx B_z(0,0) + \frac{1}{2}B_z''(0,0)z^2 - \frac{\rho^2}{4}B_z''(0,0)$$

$|z| \sim 0$ & $a \ll L$

$$\approx \mu_0 NI\left(1-\frac{2a^2}{L^2}+\frac{6a^4}{L^4}+\frac{2z^2-\rho^2}{4}\left(-48\mu_0 NI\frac{a^2}{L^4}\right)+\ldots\right)$$

$$\approx \mu_0 NI\left(1-\frac{2a^2}{L^2}+\frac{6a^4}{L^4}+12\frac{a^2\rho^2}{L^4}-24\frac{a^2z^2}{L^4}\right)$$

Radial Component $B_\rho$ :

Recall : $B_\rho(\rho,z) \approx -\frac{\rho}{2}\partial_z B_z(0,z)$

$$\approx -\frac{\rho}{2}\partial_z\left[B_z(0,0)+\frac{1}{2}B_z''(0,0)z^2\right]$$

Taylor Series App.

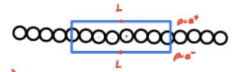
$$\approx -\rho/2\,\left(B_z''(0,0)z\right)$$

$$\approx 24\mu_0 NI\frac{a^2}{L^4}\rho z$$

- Find $B_z$ right inside the solenoid $\rho = a^-$.

$$B_z(a^-,z) \approx \mu_0 NI\left(1-\frac{2a^2}{L^2}+\frac{18a^4}{L^4}-24\frac{a^2z^2}{L^4}\right)$$

b) We draw an Amperian loop that is rectangular as in the sketch



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc} = \mu_0 NI$$

$$\approx B_z(a^-, z) - B_z(a^+, z) = \mu_0 NI$$ (we consider the $B_\rho$ is negligible $\underbrace{B_\rho(a^-,z) - B_\rho(a^+,z)}_{\approx 0}$)

$$B_z(a^+, z) = B_z(a^-, z) - \mu_0 NI$$

$$= \mu_0 NI \left[ 1 - \frac{2a^2}{L^2} + 18\frac{a^4}{L^4} - 24\frac{a^2 z^2}{L^4} \right] - \mu_0 NI$$

$$= -\frac{2\mu_0 NI a^2}{L^2} \left[ 1 + \frac{12 z^2}{L^2} - \frac{9a^2}{L^2} \right]$$

Observing the dominant term for each.

$B_z^{inside} = B_z(a^-) \sim \mu_0 NI$ $\therefore$ for $L \gg a$ we find that $B_z^{outside}$ is negligible

$B_z^{outside} = B_z(a^+) \sim \mu_0 NI \frac{a^2}{L^2} \ll 1$

c) All we need to do is find $B_z$ for $z = L/2$ by expanding.

Axial Component $B_z$

$$B_z(0, L/2) = \frac{\mu_0 NI}{2} \frac{L}{\sqrt{L^2 + a^2}} \approx \frac{\mu_0 NI}{2} \left( 1 - \frac{1}{2}\frac{a^2}{L^2} + \cdots \right)$$

$$\approx \frac{\mu_0 NI}{2}$$ leading term

Radial component $B_\rho$

$$B_\rho(\rho, L/2) \approx -\frac{\rho}{2} \partial_z B_z(0, z) \Big|_{z = L/2}$$

$$\partial_z B_z(0,z) \Big|_{z = L/2} = \frac{\mu_0 NI}{2} \left[ f'(L) - f'(0) \right]$$ Recall: $f'(u) = a^2 (a^2 + u^2)^{-3/2}$

$$= \frac{\mu_0 NI}{2} \left[ \frac{a^2}{(a^2 + L^2)^{3/2}} - \frac{1}{a} \right]$$

$$= \frac{\mu_0 NI}{2} \left[ -1/a - \frac{a^2}{L^3} \frac{1}{(1 + a^2/L^2)^{3/2}} \right]$$

$$\simeq -\frac{\mu_0 NI}{2a}$$ leading term

$$B_\rho(\rho, L/2) \simeq \frac{\mu_0 NI}{4} \left( \frac{\rho}{a} \right)$$ solving for $z = -L/2$ gives the $-$ve sign.

**5.7** A localized cylindrically symmetric current distribution is such that the current flows only in the azimuthal direction; the current density is a function only of $r$ and $\theta$ (or $\rho$ and $z$): $\mathbf{J} = \hat{\phi} J(r, \theta)$. The distribution is "hollow" in the sense that there is a current-free region near the origin, as well as outside.

**(a)** Show that the magnetic field can be derived from the azimuthal component of the vector potential, with a multipole expansion

$$A_\phi(r, \theta) = -\frac{\mu_0}{4\pi} \sum_L m_L r^L P_L^1(\cos\theta)$$

in the interior and

$$A_\phi(r, \theta) = -\frac{\mu_0}{4\pi} \sum_L \mu_L r^{-L-1} P_L^1(\cos\theta)$$

outside the current distribution.

**(b)** Show that the internal and external multipole moments are

$$m_L = -\frac{1}{L(L+1)} \int d^3x \, r^{-L-1} P_L^1(\cos\theta) \, J(r, \theta)$$

and

$$\mu_L = -\frac{1}{L(L+1)} \int d^3x \, r^L P_L^1(\cos\theta) \, J(r, \theta)$$

We work on both parts a & b by expanding the Green's Function in spherical harmonics from ch. 3.

For a current density $\vec{J}$, the vector potential for the coulomb gauge is :

$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{x}')}{|\vec{x}-\vec{x}'|} d^3x'$$

$$= \frac{\mu_0}{4\pi} \sum_{l,m} \frac{4\pi}{2l+1} \int \frac{r_<^l}{r_>^{l+1}} \vec{J}(r', \theta', \phi') Y_l^m(\theta, \phi) Y_l^{m*}(\theta', \phi') r'^2 dr' d\phi' d(\cos\theta')$$

spherical harmonics expansion.

For a cylinderically symmetric current distribution, $\vec{J} = \hat{\phi} J(r, \theta)$ :

$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \sum_{l,m} \frac{4\pi}{2l+1} \int \frac{r_<^l}{r_>^{l+1}} \hat{\phi} J(r', \theta') Y_l^m(\theta, 0) Y_l^{m*}(\theta', 0) e^{im(\phi-\phi')} r'^2 dr' d\phi' d(\cos\theta')$$

Observing the negative values of $m$ and using the identities $Y_l^{m*}(\theta, \phi) = (-1)^m Y_l^{-m}(\theta, \phi)$ & $Y_l^m(\theta, 0) = Y_l^{m*}(\theta, 0)$

$$Y_l^{-m}(\theta, 0) \, Y_l^{-m*}(\theta', 0) \, e^{-im(\phi-\phi')}$$

$$= (-1)^m Y_l^{m*}(\theta, 0) (-1)^m Y_l^m(\theta', 0) \left[\cos(m[\phi-\phi']) - i \sin(m[\phi-\phi'])\right]$$

$$= \underbrace{Y_l^m(\theta, 0) \, Y_l^{m*}(\theta', 0)}_{\text{same as +ve } m} \left[\cos(m[\phi-\phi']) \underbrace{- i \sin(m[\phi-\phi'])}_{\to 0 \text{ with +ve } m}\right]$$

Therefore, we can rewrite $\vec{A}(\vec{x})$ as :

$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \sum_{l,m} \frac{4\pi}{2l+1} \int \frac{r_<^l}{r_>^{l+1}} \hat{\phi}' J(r', \theta') Y_l^m(\theta, 0) Y_l^{m*}(\theta', 0) \cos[m(\phi-\phi')] r'^2 dr' d\phi' d(\cos\theta')$$

Let's now focus on the azimuthal part of the integral.

$$I_m \equiv \int_0^{2\pi} \hat{\phi}' \cos[m(\phi-\phi')]\, d\phi'$$

We would like to have a change of variable from primed direction $\hat{\phi}'$ to unprimed coordinates $\hat{r}, \hat{\theta}, \hat{\phi}$.

We can rewrite any arbitrary vector $\hat{x}$ in terms of a basis by:

$$\hat{x} = (\hat{r}\cdot\hat{x})\,\hat{r} + (\hat{\theta}\cdot\hat{x})\,\hat{\theta} + (\hat{\phi}\cdot\hat{x})\,\hat{\phi}$$

we have that:

$$\hat{r} = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta)$$

$$\hat{\theta} = (\cos\theta\cos\phi,\ \cos\theta\sin\phi,\ -\sin\theta)$$

$$\hat{\phi} = (-\sin\phi,\ \cos\phi,\ 0)$$

Therefore,

$$\hat{\phi}' = \sin(\phi-\phi')\sin\theta\ \hat{r} + \sin(\phi-\phi')\cos\theta\ \hat{\theta} + \cos(\phi-\phi')\ \hat{\phi}$$

Substituting $\hat{\phi}'$ in $I_m$:

$$I_m \equiv \int_0^{2\pi} \left[\sin(\phi-\phi')\sin\theta\ \hat{r} + \sin(\phi-\phi')\cos\theta\ \hat{\theta} + \cos(\phi-\phi')\ \hat{\phi}\right]\cos[m(\phi-\phi')]\, d\phi'$$

$$= \pi\left[\delta_{m,1} + \delta_{m,-1}\right]\hat{\phi}$$

by using $\int \sin(nx)\cos(mx) = 0 \ \forall n,m$

$\int \cos(nx)\cos(mx) = \pi\delta_{mn}$. & $\cos mx = \cos(-mx)$

Substituting $I_m$ back in $\vec{A}(\vec{x})$:

$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi}\sum_{l,m}\frac{4\pi}{2l+1}\int \frac{r_<^l}{r_>^{l+1}} J(r',\theta')\, Y_l^m(\theta,0)\, Y_l^{m*}(\theta',0)\, \pi\left[\delta_{m,1} + \delta_{m,-1}\right]\hat{\phi}\, r'^2\, dr'\, d(\cos\theta')$$

The $m=1$ & $m=-1$ terms are exactly the same and so we can double the $m=1$ term.

$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi}\sum_{l}\frac{4\pi}{2l+1}\, 2\pi\hat{\phi}\int \frac{r_<^l}{r_>^{l+1}} J(r',\theta')\, Y_l^1(\theta,0)\, Y_l^{1*}(\theta',0)\, r'^2\, dr'\, d(\cos\theta')$$

Writing the spherical harmonics in terms of the associated legendre polynomials.

$$\vec{A}(\vec{x}) = \frac{\mu_0}{4\pi} \sum_l \frac{4\pi}{2l+1} 2\pi \hat{\phi} \int \frac{r_<^l}{r_>^{l+1}} J(r',\theta') \sqrt{\frac{2l+1}{4\pi} \cdot \frac{1}{l(l+1)}} P_l'(\cos\theta) \sqrt{\frac{2l+1}{4\pi} \cdot \frac{1}{l(l+1)}} P_l'(\cos\theta') \, r'^2 \, dr' \, d(\cos\theta')$$

$$= \frac{\mu_0}{4\pi} \sum_l \frac{P_l'(\cos\theta)}{l(l+1)} 2\pi \hat{\phi} \int \frac{r_<^l}{r_>^{l+1}} J(r',\theta') \, P_l'(\cos\theta') \, r'^2 \, dr \, d(\cos\theta')$$

$$= \frac{\mu_0}{4\pi} \sum_l \frac{P_l'(\cos\theta)}{l(l+1)} \hat{\phi} \int \frac{r_<^l}{r_>^{l+1}} J(r',\theta') \, P_l'(\cos\theta') \, d^3x'$$

Awesome, we now have the form of the vector potential. All we need is putting the right regions.

For region inside, $r < r'$ and thus

$$A_\phi^{in}(r,\theta) = \frac{\mu_0}{4\pi} \sum_l \underbrace{\frac{1}{l(l+1)} \int \frac{J(r',\theta')}{r'^{\,l+1}} P_l'(\cos\theta') \, d^3x'}_{-m_l} \quad r^l P_l'(\cos\theta)$$

For region outside, $r > r'$ and thus

$$A_\phi^{out}(r,\theta) = \frac{\mu_0}{4\pi} \sum_l \underbrace{\frac{1}{l(l+1)} \int r'^l J(r',\theta') \, P_l'(\cos\theta') \, d^3x'}_{-\mu_l} \; \frac{P_l'(\cos\theta)}{r^{l+1}}$$